

Table 1.

MULTISYSTEM MITOCHONDRIAL DISEASES	
Organ or System Diseased	Symptoms
brain	stroke, seizures, dementia, ataxia, developmental delay
muscle	weakness, pain, fatigue
nerve	neuropathy
heart	cardiomyopathy, heart failure, heart block, arrhythmia
pancreas	diabetes, pancreatitis
eye	retinopathy, optic neuropathy
hearing	sensorineural deafness
kidney	renal failure
GI system	diarrhea, pseudo-obstruction, dysmotility

Mitochondrial diseases tend to affect multiple organ systems. The cells and organs most severely affected are those most heavily dependent on ATP, such as those listed in Table 1. Patients will frequently have multiple symptoms or signs, a circumstance that often causes confusion in diagnosis and treatment.

One of the more common presentations of mitochondrial disease in infants and young children is Leigh’s disease, first described by the pathologist Dennis Leigh in 1951. This progressive disease primarily affects the brain, with episodic deterioration that is often triggered by mild viral illnesses. Other organ systems are often involved, and there is often high blood or brain lactic acid as a result of a failure in oxidative metabolism (lactic acid is formed from glucose in the absence of oxygen). Figure 1 details the sites of metabolic defect and the percentages of cases affected in cases of Leigh’s syndrome. Complex I and IV defects are autosomal recessive diseases, with the culprit genes residing on the nuclear chromosomes. Complex V mutations are mtDNA inherited, and another 25 percent of cases are X-linked, due to pyruvate dehydrogenase deficiency (another mitochondrial enzyme, not shown in Figure 1).

One of the most common mtDNA diseases seen is due to a single point mutation at position 3,243, with an adenine to guanine mutation in a tRNA leucine gene. Patients with this mutation may have **phenotypes** ranging from asymptomatic (that is, having no visible effects) to diabetes mellitus (with or without deafness). It is estimated that 1 to 2 percent of all diabetics have the A3243G mutation as the cause, affecting 200,000 people in the United States alone. The most severe phenotype to occur from this mutation has been given the acronym MELAS, for mitochondrial encephalomyopathy, with lactic acidosis and stroke-like episodes. The variability of disease phenotype or heterogeneity of disease due to mtDNA mutations arises in part because of variations in the amount of mutated mtDNA within different tissues. This mixture of wild type and mutant DNA within a cell is called heteroplasmy. In many mtDNA diseases, heteroplasmy changes over time, so that there is an increase in mutant DNA in nondividing cells and tissues such as muscle, heart, and brain, with a decrease over time in rapidly dividing tissues such as bone marrow. SEE ALSO APOPTOSIS; DIABETES; INHERITANCE, EXTRANUCLEAR; METABOLIC DISEASE; MITOCHONDRIAL GENOME; MOLECULAR ANTHROPOLOGY.

Richard Haas

phenotypes observable characteristics of an organism

Bibliography

- Johns, D. R. "Mitochondrial DNA and Disease." *New England Journal of Medicine* 333, no. 10 (1995): 638–644.
- Raha, S., and B. H. Robinson. "Mitochondria, Oxygen Free Radicals, and Apoptosis." *American Journal of Medical Genetics* 106, no. 1 (2001): 62–70.
- Wallace, D. C. "Mitochondrial DNA in Aging and Disease." *Scientific American* 277, no. 2 (1997): 40–47.

Mitochondrial Genome

One of the defining features of eukaryotic cells is the presence of membrane-enclosed organelles. Two of these organelles, the mitochondria and chloroplast, are unique in that they contain their own genetic material necessary for proper functioning. These organelle genomes are evolutionary relics of free-living bacteria that entered into a **symbiotic** relationship with a host cell. Through the process of cellular respiration, mitochondria produce about 90 percent of the chemical energy that a cell needs to survive. The discovery that mutations in the mitochondrial genome can cause a variety of human diseases has increased our interest in this "other" human genome.

Organelle Structure and Energy Production

The mitochondria (singular: mitochondrion) are enclosed by two membranes, each a phospholipid bilayer with a unique collection of embedded proteins. The outer membrane is smooth, but the inner membrane contains extensive folds called cristae. The cristae provide a means of packing a relatively large amount of the inner membrane into a very small container, thus enhancing the productivity of cellular respiration. The number of mitochondria per cell is correlated with the cell's level of metabolic activity, with a typical cell containing hundreds to thousands of these organelles. Time-lapse photography of living cells reveals mitochondria as very dynamic structures, moving around, changing shape, and dividing.

Often described as the "power plant" of the cell, mitochondria generate ATP by extracting energy from sugar, fats, and other fuels with the help of oxygen. Mitochondria generate most of the energy in animal cells through a process called oxidative phosphorylation. In this process, electrons are passed along a series of protein complexes that are located in the inner mitochondrial membrane. The passage of electrons between these protein complexes releases energy that is stored in the membrane, and is then used to make ATP from ADP.

Mitochondrial DNA: Function and Replication

Scientists have known since the early 1960s that the nucleus is not the only location for DNA in a **eukaryotic** cell. The mitochondria (and the chloroplasts in plant cells) harbor their own small genome. The genes found on the circular 16,569 **base-pair** piece of mitochondrial DNA (mtDNA) in human cells code for thirteen proteins, two ribosomal RNAs (rRNA), and twenty-two transfer RNAs (tRNAs), all of which are essential for the production of ATP by the mitochondria. Each individual organelle contains several copies of the mitochondrial genome. Although they comprise only

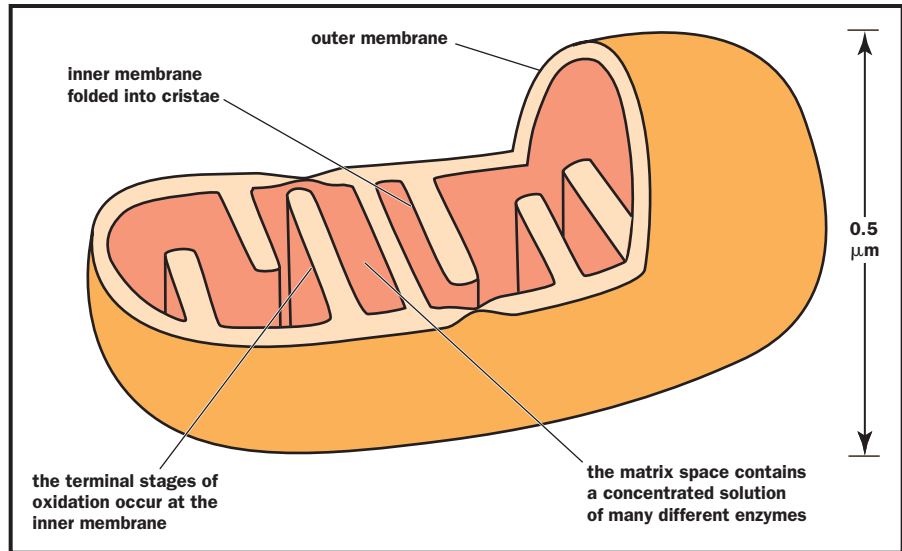
symbiotic describes a close relationship between two species in which at least one benefits

eukaryotic describing an organism that has cells containing nuclei

base-pair two nucleotides (either DNA or RNA) linked by weak bonds



Structure of a mitochondrion.



a small portion of the proteins found in the mitochondrion, all thirteen proteins encoded by the mtDNA are essential, because they are necessary for oxidative phosphorylation and the production of cellular ATP. All of the remaining mitochondrial components are encoded by nuclear genes and are imported into the organelle.

introns untranslated portions of genes that interrupt coding regions

The mitochondrial genome in mammals is extremely compact, with essentially no **introns** and very little DNA sequence between genes. Each of the protein and rRNA genes is immediately flanked by tRNA genes. Initial transcription of mtDNA produces large RNA molecules that are then processed into smaller units to generate mature tRNAs, rRNAs, and mRNAs.

The two mtDNA strands in the circular molecule can be separated based on their density (due to their differing nucleotide compositions), and are thus designated as the heavy strand (H-strand) and the light strand (L-strand). Both strands are transcribed completely, making two long RNA molecules. Since the two strands are complementary (not identical), they do not each code for the same genes. Instead, the H-strand transcript codes for most of the proteins and tRNAs, while the L-strand codes for most of the rRNAs. Ninety percent of the L-strand does not code for useful products, and is degraded after it is transcribed. The processed L-strand transcript also functions as the starting point for replication of the mitochondrial chromosome.

Endosymbiosis and Genome Reduction

Given the bacterial-like features of mitochondria and chloroplasts (small size, circular genome, and ability to divide on their own), it is believed that each organelle traces its evolutionary history to a free-living bacterial ancestor that was engulfed by a larger cell and then entered into a symbiotic relationship with the host cell. This “serial endosymbiotic theory” proposes that the evolution of the modern eukaryotic cell was a step-wise association, with the acquisition of the mitochondria preceding that of the chloroplast. The most compelling evidence for the endosymbiosis theory has come from the

analysis of complete genome sequences. Comparison of DNA sequence data has identified two specific groups of bacteria, α -Proteobacteria and Cyanobacteria, as the closest living relatives of mitochondria and chloroplasts, respectively. The mtDNA sequence information from numerous organisms has revealed remarkable similarity, reinforcing the idea of a single primary ancestor for the organelle originating very early in the evolution of the eukaryotic cell.

During the course of evolution, a large portion of mitochondrial genes were either lost or transferred to the nuclear **genome**. Elimination of genes from mtDNA is an ongoing evolutionary process made possible because their functions either become dispensable or can be replaced by nuclear functions. A comparison of the complete mtDNA sequence and the working draft of the human nuclear genome project reveals numerous areas of similarity. These regions represent mtDNA sequences that have been transferred from the **cytoplasm** to the nucleus over the course of mammalian evolution. This transfer accounts for the current nuclear location of most of the genes that encode mitochondrial proteins, including most of the proteins required for oxidative phosphorylation. Even eukaryotes that lack mitochondria (such as some protists) contain nuclear genes that encode typical mitochondrial proteins, implying that these eukaryotes once had mitochondria but subsequently lost them. SEE ALSO CELL, EUKARYOTIC; EUBACTERIA; INHERITANCE, EXTRANUCLEAR; MITOCHONDRIAL DISEASES; MOLECULAR ANTHROPOLOGY.

genome the total genetic material in a cell or organism

cytoplasm the material in a cell, excluding the nucleus

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Bibliography

Wallace, Douglas C. "Mitochondrial DNA in Aging and Disease." *Scientific American* 277, no. 2 (1997): 40–47.

"Special Section: Mitochondria." *Science* 283 (1999): 1475–1497. (A series of articles on the mitochondrial genome, mtDNA diseases, and the evolution of the organelle.)

Internet Resource

MITOMAP (a human mitochondrial genome database). Center for Molecular Medicine, Emory University. <<http://www.gen.emory.edu/mitomap.html>>.

Mitosis

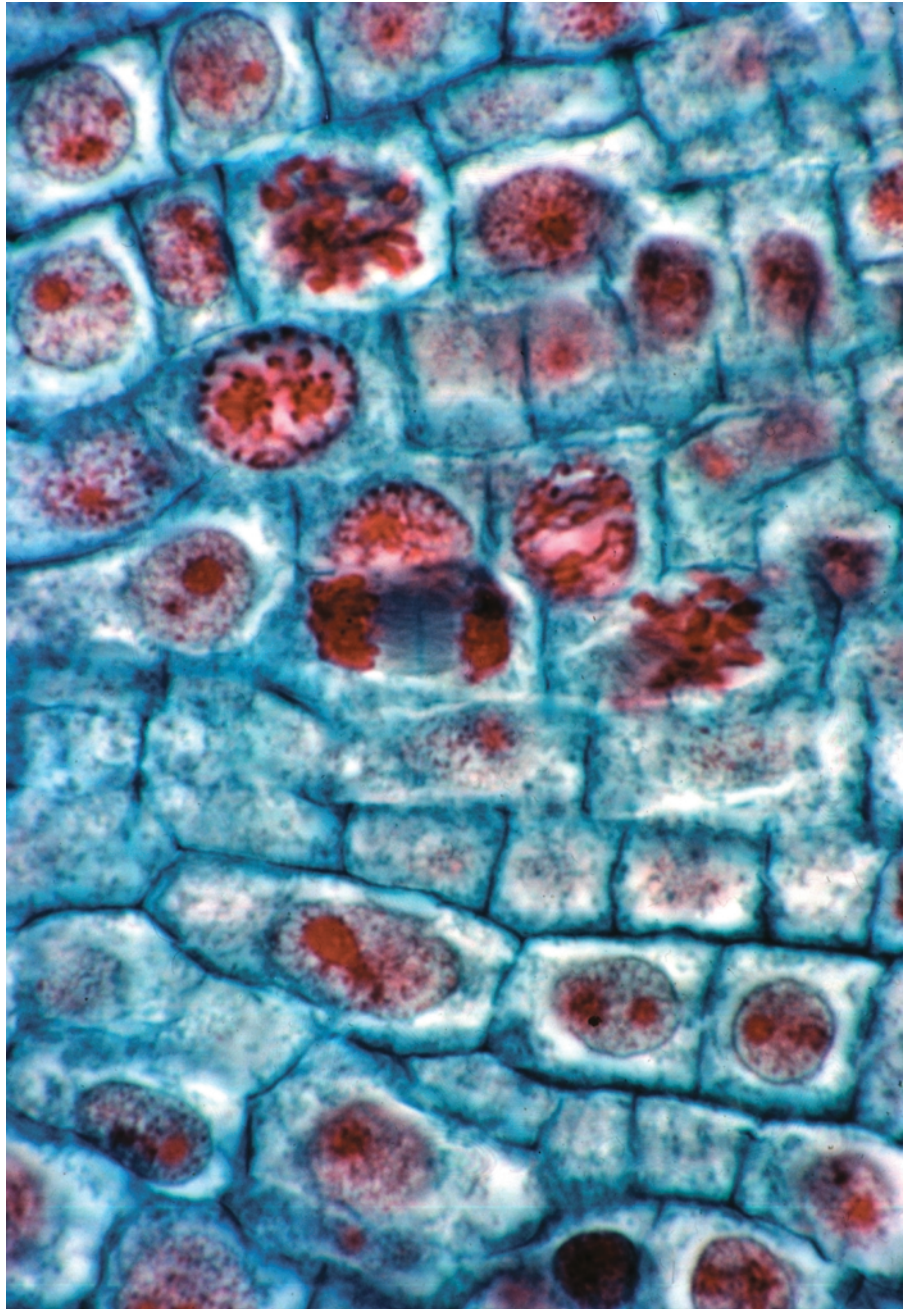
Mitosis is the process by which all cells divide. Many cells have a limited life span, and mitosis allows them to be renewed on a regular basis. Mitosis is also responsible for generating the many millions of cells that are needed for an embryo to develop into a fetus, an infant, and finally an adult.

Most human cells continually undergo a cycle of different phases. The phases have distinct names but flow smoothly into one another. The mitotic (M) phase is the phase in which the cell's genetic material is split in two. Once the phase is completed, the cell is physically divided into two daughter cells, in a process called cytokinesis.

Before entering the M phase, cells are in interphase, the phase between two cell divisions. Interphase is itself divided into three phases: G₁, S, and G₂, where G stands for gap or growth, and S for synthesis. During the G₁



Various stages of mitosis are evident in this magnification of an onion root tip. The chromosomes are stained red.



phase, daughter cells formed in the previous M phase undergo active cell growth. During the S phase, the genetic material (DNA) contained in the chromosomes is duplicated so that both of the future daughter cells receive the same set of chromosomes. This ensures that they will be genetically identical to each other and to the cell from which they originated.

somatic nonreproductive; not an egg or sperm

In human **somatic** cells, each of the forty-six chromosomes replicates to produce two daughter copies that are called sister chromatids. These two copies remain attached to each other at a single point, the centromere, which is a DNA sequence of about 220 nucleotides. The centromere has a disk-shaped protein molecule, called a kinetochore, attached to it. In interphase, the chromosomes are not visible as discrete entities under the light micro-

scope. Interphase chromosomes are uncoiled threads composed of DNA and protein molecules. This noncondensed form of chromosomes is also called chromatin.

During the G_2 phase of the cell cycle, the chromatin fibers start to condense, eventually turning into tightly coiled, compact bodies, visible as chromosomes. The cell also begins to manufacture protein fibers called microtubules, which will be used later to move the chromosomes to opposite poles of the cell, so two new daughter cells can form. Chromosome condensation and microtubule formation both begin in the G_2 phase but occur mostly during the first stage of the M phase, which is called prophase. The microtubules are organized into a three-dimensional spindle apparatus, where each fiber of the spindle apparatus connects one cell pole to the other like a bridge.

During the next stage of mitosis, called prometaphase, the envelope surrounding the cell nucleus breaks down so that the chromosomes are free to migrate to the central plane of the spindle apparatus. A second group of microtubules grows out, to connect the two opposite sides of the kinetochore to the two poles of the spindle. This arrangement is crucial for ensuring that the two sister chromatids end up in two separate daughter cells rather than being pulled into the same cell.

In the next stage of mitosis, metaphase, the chromosomes become maximally condensed and line up in an imaginary plane, called the metaphase plate, in the center of the cell and perpendicular to the spindle apparatus. All the centromeres are neatly arranged in a circle, about halfway between the two cell poles.

In human cells, at this point the twenty-three pairs of chromosomes, each made up of two condensed sister chromatids held together by a centromere, are visible under the microscope. Unlike in meiosis, the paternal and maternal copies in each pair of chromosomes align independently in the metaphase plate and are not associated with each other. At the end of metaphase, the centromeres that hold the two sister chromatids together all divide simultaneously.

During the next stage, anaphase, microtubules that are attached to the sister chromatids' kinetochores draw the chromatids quite rapidly to opposite poles of the spindle. The separation of sister chromatids completes the partitioning of the replicated genetic material.

The only task remaining during the final phase of mitosis, telophase, is to disassemble the spindle apparatus and re-form the nuclear envelope around each set of sister chromatids. The chromatids can be called chromosomes again, because they each have their own centromere. The chromosomes begin to uncoil, and the genes they carry begin to be expressed again. This is the end of mitosis.

The cell cycle is completed by cytokinesis, the physical division of the cytoplasm into two daughter cells. By the time cytokinesis occurs, other cytoplasmic organelles, such as mitochondria, already have been replicated during the S or G_2 phases, and they have also been directed to the areas around the cell poles that will become the daughter cells. Cytokinesis is followed by the G_1 phase, with active cell growth occurring in each of the two daughter cells.



Differences between Mitosis and Meiosis

Mitosis occurs in all eukaryotic cell tissues and produces genetically identical daughter cells with a complete set of chromosomes. In humans, mitosis produces somatic cells that are diploid, which means they contain two non-identical copies of each of the twenty-three chromosomes. One copy is derived from the person's mother and the other from the person's father.

Meiosis, on the other hand, occurs only in testis and ovary tissues, producing sperm and ova (eggs). The gametes that are produced by meiosis in humans are haploid, containing only one copy of each of the twenty-three chromosomes. Because of recombination and independent assortment of parental chromosomes, the daughter cells produced by meiosis are not genetically identical.

In mitosis, one round of DNA replication occurs per cell division. In meiosis, one round of DNA replication occurs for every two cell divisions. Prophase in mitosis typically takes about thirty minutes in human cells. Prophase in meiosis I can take years to complete. SEE ALSO CELL, EUKARYOTIC; CELL CYCLE; MEIOSIS; NUCLEUS; REPLICATION.

Silke Schmidt

Bibliography

Nasmyth, Kim. "Segregating Sister Genomes: The Molecular Biology of Chromosome Separation." *Science* 297, no. 5581 (2002): 559.

Raven, Peter H., and George B. Johnson. *Biology*, 5th ed. New York: McGraw-Hill, 1999.

Model Organisms

A model organism is a species that biologists choose to study, not necessarily because it has any inherent medical, agricultural, or economic value, but because it has certain traits that make it easy and convenient to work with. Studying model organisms enables researchers to perform experiments that might be impossible to carry out, due to logistical, financial, or ethical constraints, on organisms of more practical interest, such as humans.

This approach has been tremendously successful in the fields of genetics and molecular and cellular biology because, at their most fundamental levels, biological processes are remarkably similar across species. For example, the genetic code and much of the cellular machinery responsible for **replication**, **transcription**, **translation**, and gene regulation are essentially identical in all eukaryotic organisms. In many cases, genes have even been demonstrated to be functionally interchangeable between humans and baker's yeast.

Among the most commonly studied model organisms are: *Escherichia coli* (a bacterium), *Saccharomyces cerevisiae* (baker's yeast), *Dictyostelium discoideum* (slime mold), *Drosophila melanogaster* (a fruit fly), *Caenorhabditis elegans* (a soil roundworm), *Brachydanio danio* (zebrafish), *Xenopus laevis* (African clawed frog), *Arabidopsis thaliana* (a mustard weed), *Zea mays* (maize, or corn), and *Mus musculus* (mouse). Others include sea slugs, sea

replication duplication of DNA

transcription messenger RNA formation from a DNA sequence

translation synthesis of protein using mRNA code



This albino African clawed frog is one of the most commonly studied model organisms.

urchins, cyanobacteria, *Chlamydomonas* (an alga), puffer fish, *Tetrahymena* (a protozoan), and rats.

Useful Traits and Attributes

Most model organisms share a set of common features that make them amenable to study in the laboratory: They are generally small, easy, and inexpensive to rear in the lab, and reproduce quickly and prodigiously. In addition, the best genetic model organisms have small genome sizes, and many can reproduce sexually, allowing researchers to cross-breed individuals of different genotypes.

Beyond these common traits, most model organisms have one or several unique attributes that make them ideal for a particular line of research. For instance, zebrafish readily produce many large, transparent embryos, and are therefore a favorite research subject for developmental biologists. The roundworm, *Caenorhabditis elegans*, has a simple but nonetheless sophisticated nervous system, and displays simple behaviors, such as movement, feeding, and mating. These properties make it well suited for neurobiology and behavioral genetics.

The Model Mouse

Besides certain primates (such as monkeys and chimpanzees), which are costly and difficult to rear in the laboratory, the model organism most closely related to humans is the mouse, *Mus musculus*. The mouse genome is about the same size as the human genome, and the organization of genes (the order of genes on chromosomes) is strikingly similar between the two species. Findings from the nearly complete mouse genome sequencing project indicate that mice and humans share about 95 percent DNA sequence similarity. This means that any gene in humans is likely to have an identical or very similar counterpart (homologue) in the mouse genome.



homologous carrying similar genes

In addition, it is much easier and less expensive to study genes in mice. A technology that has made the mouse an invaluable genetic model system is the ability to engineer “knockout” strains. These are mutant strains in which a single known gene has been selectively deleted from the genome of every cell. For human genes implicated in diseases, knocking out the homologous gene in mice can provide an excellent model system for studying the disease. The knockout mouse may show disease conditions similar to those of the human disease. Learning how the elimination of the gene in the mouse contributes to the mouse disease may then give important clues about the involvement of the **homologous** gene in the human disease. “Disease model” mouse strains are available for such disorders as cancer, Alzheimer’s disease, arthritis, diabetes, heart disease, cystic fibrosis, and obesity. **SEE ALSO** *ARABIDOPSIS THALLANA*; *ESCHERICHIA COLI*: (*E. COLI* BACTERIUM); FRUIT FLY: *DROSOPHILA*; MAIZE; RODENT MODELS; ROUNDWORM: *CAENORHABDITIS ELEGANS*; TRANSGENIC ANIMALS; YEAST; ZEBRAFISH.

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Bibliography

Alberts, Bruce, et al. *Molecular Biology of the Cell*, 3rd ed. New York: Garland Publishing, 2002.

Pines, Maya, ed. *The Genes We Share with Yeast, Flies, Worms, and Mice*. Chevy Chase, MD: Howard Hughes Medical Institute, 2001. (Available from the Howard Hughes Medical Institute Web site: <www.hhmi.org>.)

Watson, James D., et al. *Recombinant DNA*, 2nd ed. New York: Scientific American Books, 1992.

Internet Resources

euGenes: Genomic Information for Eukaryotic Organisms. <<http://iubio.bio.indiana.edu>>.

WWW Virtual Library: Model Organisms. <<http://ceolas.org/VL/mo/>>.

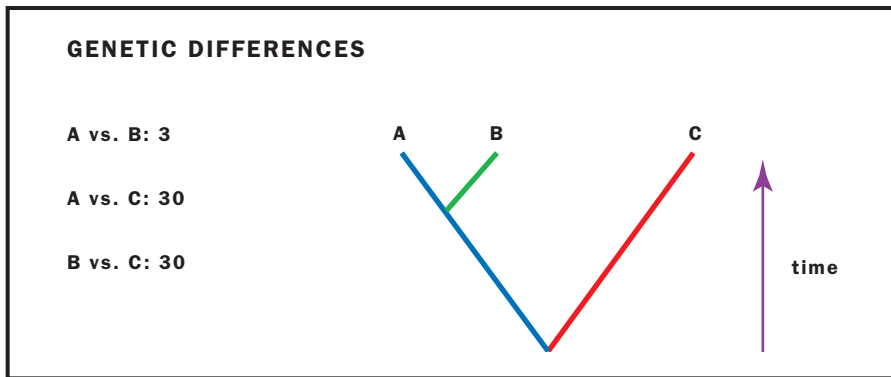
Molecular Anthropology

Anthropology is the study of the origin and development of the human species. Molecular anthropology uses the tools and techniques of molecular genetics to answer anthropological questions, especially those concerning the origins and spread of humans across the globe. These questions mainly fall under the heading of physical or biological anthropology, as opposed to cultural anthropology, which studies social relationships, rituals, and other aspects of culture.

Tracing Human Origins through Genetic Data

Molecular anthropology attempts to answer such questions as whether humans are more genetically similar to chimpanzees than to gorillas; in what region or regions modern humans first developed; what the patterns are of migration and mixture of early human populations; and whether Neanderthals were a different species, and whether they died out or mixed in with modern humans. Molecular anthropology is perhaps best known for the studies that surround the discovery of “mitochondrial Eve” (discussed below), although the meaning of that discovery is often misunderstood.

Two major approaches are used in addressing these questions, both of which involve analyzing DNA. The first and most common approach is



The fundamental postulate of molecular anthropology is that closer genetic similarity indicates a more recent common ancestry. Here, the similarity of A and B vs. C indicate A and B are more closely related, as shown in the tree diagram.

to compare the DNA of groups of *living* organisms, for example, comparing humans to humans or humans to primates. The second approach relies on isolating and analyzing DNA from an *ancient* source, and comparing it to other ancient DNA or to modern DNA. In both cases, the number of differences between the DNA sequences of the two groups are determined, and these are used to draw conclusions about the relatedness of the two groups, or the time since they diverged from a common ancestor, or both.

The results of molecular anthropological studies are rarely used alone. Instead, the data are combined with information from fossils, archaeological excavations, linguistics, and other sources. Sometimes the data from these different sources conflict, however, and much of the controversy in anthropology centers around how much weight to give each when this occurs.

Advantages of DNA Comparisons

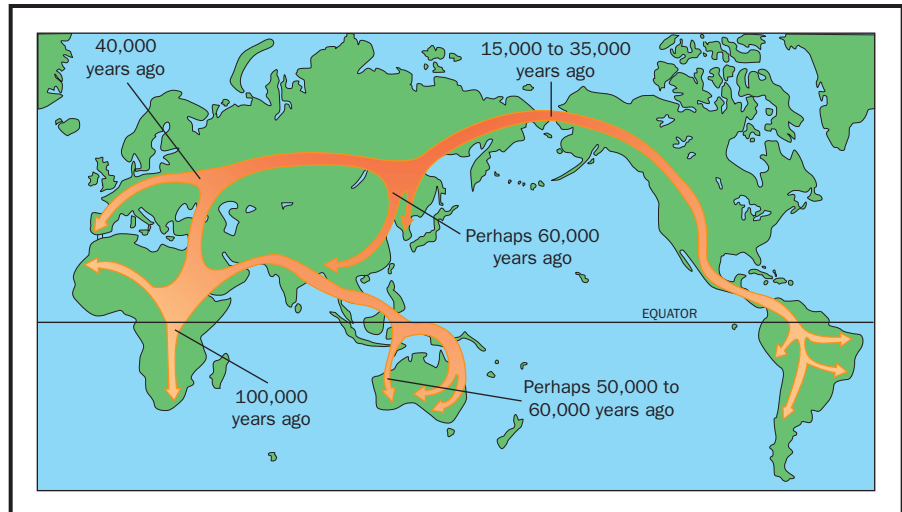
The essential postulate on which molecular anthropology is based is that closer genetic similarity indicates a more recent common ancestry. All organisms are believed to have evolved from a single ancestor. As different life forms evolved, their DNA began to diverge through the processes of mutation, natural selection, and genetic drift. Even within the same species, populations that do not interbreed will accumulate genetic differences, which increase over time. The number of these differences is proportional to the amount of time since the two groups diverged.

There are several advantages to comparing DNA data instead of external physical characteristics (collectively called the phenotype). Environmental factors can shape the phenotype to make two individuals with the same genetic makeup look different. For instance, nutrition has a profound effect on height, and if we used average height to classify humans, we might mistakenly conclude that medieval humans represented a different subspecies because they were significantly shorter than modern humans. DNA comparisons, on the other hand, would show no significant difference between these groups.

Another advantage is that DNA sequence differences can be easily quantified—two base changes in a gene are more different than one. Despite being random events, **mutations** occur at a fairly steady rate, constituting a “molecular clock,” and so the number of differences can be used to estimate the time since the two organisms shared a common ancestor.

mutations changes in DNA sequences

Molecular anthropologic data has been used to argue that modern humans arose in Africa and migrated out to replace archaic populations elsewhere. The details and the timing are still controversial.



Finally, since all organisms contain DNA, the sequences of any two organisms can be compared. The same techniques used in molecular anthropology can also be applied to evolutionary questions in other species, to determine the evolutionary relations between different animal species, for instance, or even between bacteria and humans.

Caveats About Sequence Comparisons

On the other hand, the simplicity and power of sequence comparisons can lead too easily to an oversimplified interpretation of results, and to conclusions that may sound more significant than they are. A prime example is the often-repeated statement that humans and chimpanzees share 98 percent of their DNA.

It may be true that 98 out of 100 bases are the same in the two genomes, but what is the significance of this fact? It does not mean that 98 percent of our genes are identical. In fact, almost all of them differ slightly, some dramatically. It also does not tell us whether the significant differences between humans and chimps arise from a few very different genes, or many slightly different ones. Moreover, there are significant differences in genome structure not accounted for by the sequence comparison. For instance, humans have forty-six chromosomes, whereas chimpanzees have forty-eight; they have about 10 percent more DNA than humans do; and humans have more copies of a certain kind of **transposable genetic element** than they do.

Most importantly, the sequence similarity certainly does not tell us that humans “are” 98 percent chimpanzee—we are two entirely different species, as is obvious from differences in anatomy and behavior. If the profound differences between humans and chimps are not reflected in the sequence data, it may be that this simple tabulation of difference does not adequately summarize the ways in which DNA can cause two organisms to differ.

The 98 percent figure, therefore, may be used to say that chimps and humans are closely related, and are more closely related to each other than either is to an organism with a greater number of sequence differences, such

transposable genetic element DNA sequence that can be copied and moved in the genome

as the orangutan. However, it may not be used to draw conclusions about the similarity of humans and chimps as organisms.

Types of DNA Comparisons

The human genome is much too large to sequence all of it to make comparisons, using current technology. Instead, much smaller portions of it are used. One strategy is to compare gene sequences, such as the sequence for hemoglobin. A potential problem with this is that most mutations in such useful genes are harmful, and so the few harmless mutations they accumulate may be similar between two individuals, despite a long evolutionary separation. Nonetheless, gene comparisons are useful for distantly related species, such as humans and yeast.

An alternative is to look at noncoding regions of DNA. These include microsatellite DNA sequences, a type of repetitive DNA element found throughout the genome. Because these sequences do not code for protein, most mutations in them do not affect the viability of the organism in which they occur. Thus they accumulate mutations more quickly. Another option is single nucleotide polymorphisms. These are sequences which differ among individuals or groups by a single nucleotide. There are millions of such sequences in the genome. Because there are so many different forms, these noncoding sequences are especially useful for determining kinship among closely related individuals, such as members of a tribe or extended family.

One potential problem with sequence comparisons is back mutation, in which a base mutates to another, and then reverts to the original (for example, $C \rightarrow T \rightarrow C$). When this occurs, two sequences may appear to be more closely related (less separated in time) than they really are, since the intervening mutation (the change from C to T, in this case) may not be apparent. Because of back mutation, the observed number of differences between sequences represents the minimum actual difference. Correction factors can be applied to estimate the true difference.

Another potential problem with any sequence on a chromosome, whether or not that sequence codes for a protein, is that most chromosomes do not remain intact during **meiosis**. This is because crossing over occurs, in which homologous chromosomes recombine (exchange segments). After a few generations, it becomes very difficult to track individual sequences and compare them with any confidence to similar sequences in another person. To avoid this problem, molecular anthropologists focus on two sources of DNA that do not recombine: the Y chromosome and mitochondrial DNA.

The Y Chromosome

The Y chromosome, which determines male sex, does not undergo recombination along most of its length. Instead, it passes intact from father to son. A man's Y chromosome, therefore, is a more-or-less exact copy of the one possessed by his father, grandfather, great-grandfather, and so on back through time. Like any other DNA segment, it may mutate, and any changes it accumulates are faithfully passed along as well. Two brothers are likely to have exactly the same Y chromosome sequence. Two men whose last common male ancestor was ten generations ago, however, are likely to have slightly different sequences. Comparison of the sequences of two Y chromosomes, therefore, can show how closely related two males are.

meiosis cell division that forms eggs or sperm

Analysis of Y chromosome microsatellite sequences was used to show that Thomas Jefferson was an ancestor of some of the male descendants of Sally Hemings, a slave owned by Jefferson.

Skulls from a Neandertal (left) and an anatomically modern human Cro-Magnon (right). Molecular anthropologists analyze modern and ancient DNA to determine how the Neandertal is related to modern humans.



Y chromosome analysis has been used to track migration of human populations, and to study the relatedness of modern populations. For instance, Jews and Palestinian Arabs derive from a common ancestral population that lived in the Middle East about 4,000 years ago. Recent studies have linked the ancestors of American Indians to several small populations in Siberia, confirming the predominantly Asian origin of American Indians and refining the understanding of their migration history. Many other similar studies have been performed, providing an increasingly clear (and complex) picture of human migration and mixture.

Mitochondrial DNA and the Origin of Modern Humans

Mitochondria are energy-harvesting organelles in the cell. They are inherited only from the mother, and so track maternal inheritance in the same way that the Y chromosome tracks paternal inheritance. Like microsatellite DNA, mitochondrial DNA accumulates mutations faster than chromosomal coding DNA.

One of the earliest and most famous mitochondrial studies was used to address a central question in anthropology: Where and how did modern humans originate?

The *Homo* genus itself is universally believed to have originated in Africa. Groups of *Homo erectus* are known to have migrated out of Africa, populating Europe and Asia between one and two million years ago. *H. erectus* gradually changed in character, so that by about half a million years ago, it had taken on some more modern characteristics. Anthropologists call these groups “archaic” modern humans. They include the Neandertals, who lived in Europe and the Middle East from 150,000 to 28,000 years ago. Did modern humans evolve from these older populations in several different regions simultaneously? Or did they arise from a small group in Africa, and spread out from there? If so, did they mix with less advanced local populations (such as Neandertals), or replace them entirely?

The scientists who performed the mitochondrial DNA study (Rebecca Cann, Mark Stoneking, and Allan Wilson) reasoned that populations that had been in one place for only a short period of time would show very little variation in their mitochondrial DNA, since they all shared a relatively recent common ancestor. This would be the case in a modern human population if it had only recently migrated into the area in which it is found. (Such relative genetic homogeneity in newly formed populations is known as the founder effect.) In contrast, populations that have remained in place for long periods have much more ancient common ancestors, and therefore have more mitochondrial DNA variations.

To perform their analysis, the scientists collected samples from different ethnic groups from all over the world. They found that the populations with the greatest amount of sequence variation were in sub-Saharan Africa, indicating these were the groups with the most ancient ancestry. All other groups had much less variation, indicating more recent arrivals of those groups in those regions. Cann, Stoneking, and Wilson went on to estimate the date at which all these groups had their most recent common ancestor. Using a figure of 2 to 4 percent sequence divergence per million years, they estimated that the most recent common ancestor lived approximately 200,000 years ago.

The simplest explanation, they argued, was that ancestors of the non-African *Homo sapiens* migrated out of Africa about 200,000 years ago to populate other regions, over time replacing the nonmodern humans (*H. erectus*, Neandertals, and possibly others) already living in these regions. They argued that the relatively short time since the divergence of all modern humans was too brief to support the alternative hypothesis, that each local group of archaic humans had independently evolved modern traits, a model called multiregional evolution.

The conclusions drawn in this study are still controversial. Numerous other studies have been done since, and the data have been subjected to multiple different analyses. Some studies suggest differing dates for the most recent common ancestor (ranging from 100,000 to 400,000 years ago), and others suggest that an exclusive African origin is not the only possible interpretation of the data.

It is important to keep in mind that the vast number of comparisons that must be made in such studies require computer programs, not only to make the comparisons, but to draw from them the simplest “family tree” that fits. Much of the controversy surrounds the assumptions that must be built into these programs in order to generate results. The mutation rates by which events are timed (the “molecular clock”) are also not known with precision, leading to further uncertainties about the exact timing of migrations.

Mitochondrial Eve

In their study, Cann, Stoneking, and Wilson pointed out that the patterns of mitochondrial variations they saw suggested that all the mitochondria of all living groups could be traced back to a single woman who lived in Africa approximately 200,000 years ago. Many people at the time of the original study and since have misinterpreted the results to claim there was a single female ancestor for all modern humans, dubbed “Eve.”



It is true the study showed that the mitochondrial DNA in all living humans probably derives from this single woman. However, our nuclear DNA certainly does not derive exclusively (perhaps even at all) from this woman, and the thirty thousand or more genes in our nuclear DNA are far, far more important in determining our characteristics than the thirty-seven mitochondrial genes. Because of recombination, our nuclear DNA cannot be traced back to any single person. Rather, it is an amalgam of countless ancestors through time.

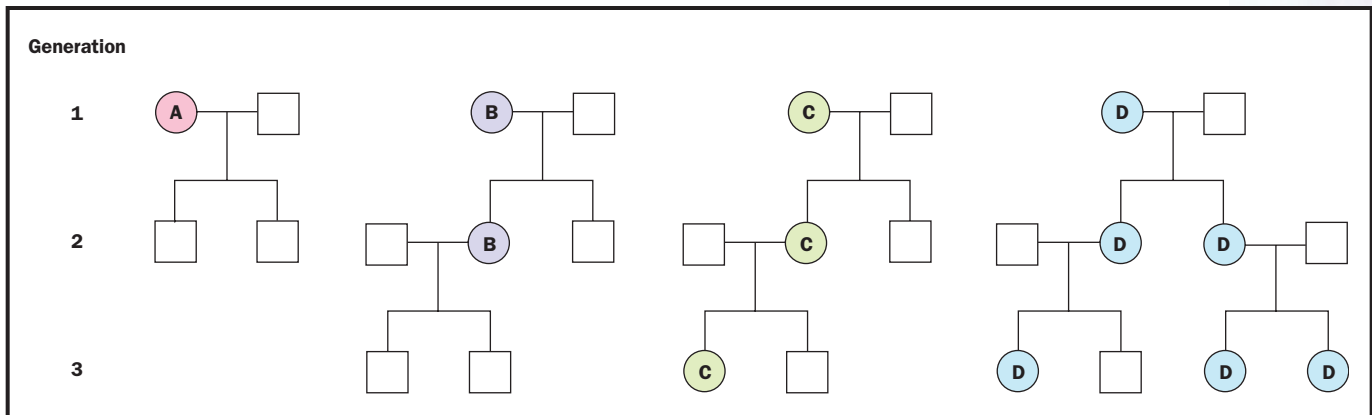
Mitochondrial Eve was also not the first modern human woman, nor the only woman in existence at the time she lived. She was not even the only woman in her local population; it is estimated that Eve was one of about 10,000 people in her population. There was really nothing particularly special about her, except that, by chance, the descendants of her mitochondria happen to have ended up in the cells of every living human. Even this, which sounds remarkable, is pretty much what we should expect from small populations.

To understand why, consider four couples, each of which has two children. Remember that mitochondria are passed from the mother to each child. One couple has two boys. Each boy inherits the mother's mitochondria, but neither passes them on to his children. The mother's mitochondrial type thus becomes extinct in one generation. Two of the couples have a boy and a girl, while the fourth has two girls. These four daughters go on to have children of their own, each with the same distribution according to sex. Whenever a family has only boys, a mitochondrial type becomes extinct. Any time a family has only girls, the mitochondrial type handed down from the mother becomes more common in the next generation. In a small population, over time, it is highly likely that one type will become most prevalent, ultimately becoming the one type found in all the members of the population. Looking back, we would give the name "Eve" to the original mother of that line of mitochondrial genetic inheritance.

A similar phenomenon occurs with the Y chromosome, for exactly the same reasons: Any family with only girls extinguishes that Y chromosome type. The "Y chromosome Adam" lived 60,000 to 150,000 years ago. There is no reason to expect that "Y chromosome Adam" would know "mitochondrial Eve"; indeed, even without the dates to make it impossible, it would be a remarkable coincidence if they had.

Neandertal DNA

DNA can be extracted from some archaeological samples, allowing direct sequencing and comparison with modern DNA. This has so far been possible with specimens up to about 40,000 years old (the dating of such samples is often inexact). DNA is isolated, purified, amplified with the polymerase chain reaction, and sequenced. By this technique, DNA from extinct animals such as the woolly mammoth has been obtained, but not dinosaur DNA, which is millions of years old. The DNA that can be isolated is typically highly fragmented and incomplete, and unsuitable for cloning the whole organism. One application is to analyze the DNA from plant and animal material at camp sites to determine the diet of early humans.



DNA can also be extracted from ancient human remains. As of summer 2002, mitochondrial DNA from two Neandertal skeletons had been extracted, sequenced, and compared. The first was from Germany, and was approximately 35,000 to 70,000 years old. A 378-base pair sequence was determined, and compared to almost one thousand different modern humans. On average, it differed at twenty-seven locations, while modern humans differed among each other at an average of only eight locations. There was some overlap, however, with the least number of differences between Neandertals and modern humans being twenty-two, and the greatest difference noted between modern humans being twenty-three.

The second skeleton was from Russia, and was 29,000 years old. A 345-base pair sequence was determined. It differed at twenty-three locations from a standard modern human sequence, but at only twelve locations compared to the German Neandertal DNA.

Keeping in mind that only two Neandertal sequences have been studied so far, some tentative conclusions have been offered from these data. The amount of difference between the two Neandertal sequences is similar to the amount found between randomly selected modern humans, suggesting that these two specimens, despite being separated by thousands of years, were indeed part of the same lineage.

The amount of difference between the Neandertal skeletal DNA and modern humans suggests that Neandertals were genetically distinctly different from modern humans in their mitochondrial DNA. Were they different enough to constitute a separate species? That is much less clear, and is a source of disagreement among anthropologists. The difference is much less than that between modern humans and chimpanzees, for instance, which suggests that they were not separate species, but it is greater than the differences among subspecies of chimpanzees, which suggests that perhaps they were. Scientists have not been able to compare Neandertal sequences to sequences from anatomically modern humans living at the same time as the Neandertals. It may be that those sequences would be more similar. At present, the relationship of Neandertals to modern humans has still not been conclusively determined.

Conclusion

Using the tools of molecular genetics, DNA sequences can be compared among groups to test hypotheses about the evolutionary relatedness of

Mitochondrial inheritance is through the mother (circles) only. After only three generations, mitochondrial type D is present in 75 percent of the women in this hypothetical population. Over time, it is quite likely that only one type will be left, simply by chance.

organisms, and about the time that has elapsed since divergence. Molecular anthropology has made major contributions to understanding the migration and mixture patterns of human groups. It has also provided significant new insights into the rise and spread of modern humans and their relation to earlier human groups. As more data becomes available and better models are devised for their interpretation, the results are likely to become less provisional and more certain. SEE ALSO INHERITANCE, EXTRANUCLEAR; FOUNDER EFFECT; MITOCHONDRIAL GENOME; MUTATION RATE; POLYMORPHISMS; POPULATION BOTTLENECK; REPETITIVE DNA ELEMENTS; SEQUENCING DNA; Y CHROMOSOME.

Richard Robinson

Bibliography

- Avise, John C. *Molecular Markers, Natural History, and Evolution*. New York: Chapman and Hall, 1994.
- Hammer, M. F., et al. "Jewish and Middle Eastern Non-Jewish Populations Share a Common Pool of Y Chromosome Biallelic Haplotypes." *Proceedings of the National Academy of Sciences* 97, no. 12 (2000): 6769–6774.
- Marks, Jonathan. *What It Means to Be 98% Chimpanzee: Apes, People, and Their Genes*. Berkeley, CA: University of California Press, 2002.
- Relethford, John H. *Genetics and the Search for Human Origins*. New York: Wiley-Liss, 2001.
- Vigilant, L., et al. "African Populations and the Evolution of Human Mitochondrial DNA." *Science* 253 (1991): 1503–1507.

Internet Resource

Y Chromosome Links. <<http://john.hynes.net/y.html>>.

Molecular Biologist

Biologists study life on many different levels. For example, a cellular biologist is concerned with the most basic unit of life, the single cell, whereas an evolutionary biologist may investigate the origin and genealogical history of a particular species of plant or animal life. The molecular biologist is concerned with understanding the biological phenomena of life at the molecular level. Molecular biology is a multifaceted discipline of recent origin, having emerged in the 1980s from the related fields of biochemistry, genetics, and cell biology.

Basic and Applied Research

A molecular biologist might investigate the genetic basis of a disease, analyzing the gene or genes suspected of causing the disease at the molecular level by using the biochemical technique of DNA sequencing. Genes code for proteins; that is, a particular gene contains the molecular information for producing one particular protein. A gene is expressed through the process of **transcription**. The DNA of the gene is transcribed by a protein known as RNA polymerase. Some genes are expressed frequently, others rarely or only during special times in development. Thus, a molecular biologist might also seek to understand the regulation of gene expression by studying how and when a gene's RNA message (mRNA) appears.

The mRNA resulting from gene expression is the blueprint for the protein. Ribosomes, the cell's protein synthesis factories, translate the mRNA

transcription messenger RNA formation from a DNA sequence



(read the message) and assemble the protein. After **translation**, a protein may be modified by covalent attachment of carbohydrates and lipids to particular amino acids. A molecular biologist might seek to determine the three-dimensional structure of a modified protein using techniques like X-ray diffraction and nuclear magnetic resonance.

In addition to carrying out basic research, molecular biologists may also work in applied research. Using recombinant DNA technology, for example, molecular biologists have created economical vaccines against deadly diseases. The molecular biologist often works at the frontier or cutting edge of a discipline. The rewards of such work include the thrill of intellectual discovery and the opportunity to conduct independent research. Also, the efforts of molecular biologists can bring great benefits to society.

Career Preparation

To prepare for a career as a molecular biologist, the student should begin by taking a broad selection of science and math courses in high school. Typically, such a course of study would include biology, chemistry, and physics, as well as geometry, algebra, and calculus. Good communication skills are

At the Los Alamos National Laboratory, molecular biologist Susana Delano performs an analysis of anthrax DNA utilizing gel electrophoresis. This New Mexico lab has performed research and DNA analysis to aid investigations of anthrax exposures in the United States.

translation synthesis of protein using mRNA code

very important for a scientist, so English and public speaking should not be neglected. At college, the student may wish to pursue a field of study involving biology or chemistry as a major, with an emphasis on laboratory training.

Usually about four to five years of study are needed to satisfy the requirements for an undergraduate (bachelor's) degree. Time for specialization comes at the graduate level. At this point, students should begin to focus their curiosity and choose a major area of interest, such as biochemistry, biophysics, genetics, or cell biology. They should also begin to look for a university that has an excellent reputation for research in the area of choice. Then, the student must seek out a particular laboratory and research advisor for real hands-on training and experimentation.

Two degree tracks are typically offered in graduate school: a master's and a doctoral (Ph.D.) program. On average, a student may earn a master's degree in molecular biology in roughly three years; earning the Ph.D. degree may require four to six years of work. Many pharmaceutical and biotechnology companies actively recruit scientists at the master's level who have training in molecular biology techniques. Students who have obtained a Ph.D. degree in molecular biology commonly undertake postdoctoral training: two to four years of additional study and research. Traditionally, only students interested in academic or government careers pursued postdoctoral studies; however, today many private companies offer one- and two-year postdoctoral positions as a means of attracting top scientific talent.

The annual salaries of molecular biologists can range from \$20,000 to \$150,000 or more, and are influenced by many factors, such as education (master's versus doctoral degree), experience (just beginning or a seasoned veteran), field of expertise ("hot" fields pay better), employer location (big city or small town), and the local supply of and demand for trained life scientists. Typically, industry positions come with somewhat higher salaries than academic or government positions; however, job security in industry may be tied to the financial success of the company. Academic and government positions may offer more intellectual independence, but sometimes lower salaries. The demand for well-trained, creative molecular biologists in government, industry, and academia continues to grow as our knowledge of life's basic processes deepens. SEE ALSO GENETICIST; LABORATORY TECHNICIAN.

Samuel E. Bennett and Dale Mosbaugh

Molecular Clock *See Mutation Rate*

Morgan, Thomas Hunt

Geneticist
1866–1945

Thomas Hunt Morgan proved the validity of the chromosomal theory of heredity and led a research group whose insights into the physical nature of inheritance propelled genetics into the center of biology in the twentieth century.

Training and Early Interests

Morgan was born and raised in Kentucky, and received his bachelor's degree from the State College of Kentucky in 1886. He pursued graduate study at

Johns Hopkins University in Baltimore, and eventually became a professor of biology at Bryn Mawr College in 1891. His early interests were in developmental biology and evolution. After moving to Columbia University in 1901 and coming under the influence of the great cell biologist Edwin Wilson, Morgan turned his attention to understanding the physical basis of inheritance, which he saw as a means to test theories about the role of mutation in evolution.

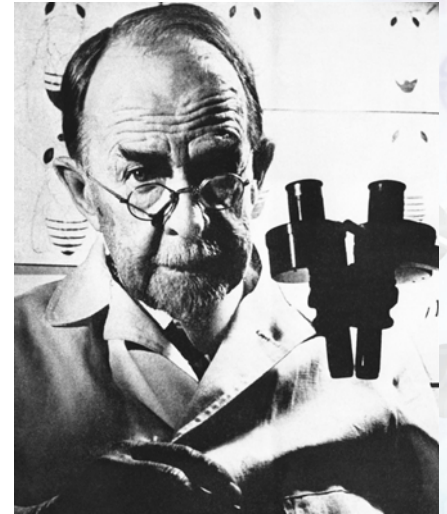
At the time Morgan began his work, chromosomes had been seen in cells, but their significance was unknown and not widely considered. A student of Wilson's, Walter Sutton, had recently proposed that chromosomes carried the genetic material, but had little evidence to support this important hypothesis. At the time, the gene itself was an abstract concept with no known physical correlate, and many scientists thought it was not a physical entity at all, but only a convenient fiction for describing some experimental results. In fact, it was Morgan's use of the term "gene" that helped bring it into general use in science.

To attack the issue of heredity, Morgan chose to work with the fruit fly, *Drosophila melanogaster*. This fly requires little space, breeds quickly, has many observable characteristics, and has only four chromosomes, making it an ideal model organism for genetics studies. Morgan also gathered a trio of very bright students, Hermann Muller, Alfred Sturtevant, and Calvin Bridges, and cultivated an egalitarian system of collaboration that was unknown in most other labs. The combination of the right question, the right model, the right collaborators, and some luck allowed Morgan and his group in their lab, dubbed "The Fly Room," to make their fundamental discoveries. Beginning in 1908, they proved that chromosomes do indeed carry the genes, that genes are discrete physical things arranged on chromosomes like beads on a string, that genes change places on chromosomes, that genes can be mutated and those mutations are faithfully inherited, and that mutations can be caused by exposure to high-energy radiation or other environmental phenomena.

A Lucky Discovery

This long string of seminal discoveries began with the discovery of a single male white-eyed fly among the many thousands of normal red-eyed ones. Morgan bred this mutant male with a red-eyed female. All the offspring were red-eyed, indicating the white form of the gene (called the white **allele**) was "recessive" to the dominant red allele: Flies carried the mutant allele, but its effects did not show up. When these offspring were crossed, the ratio of red to white was 3:1, just what would be expected for a classical **recessive** trait.

However, Morgan noted an unusual fact about the white-eyed flies—all of them were male. Morgan knew that the female *Drosophila* had two so-called X chromosomes, while the male had only one. Combining this fact with his discovery that only males showed the white-eye trait, he reasoned that the white-eye mutant allele must be on the X chromosome. Males show the white-eye trait because the mutant white allele is the only one they have—they don't have a second X chromosome with a normal red allele. Females rarely show the white-eye trait, because they have a normal red-eye allele on the other X chromosome.

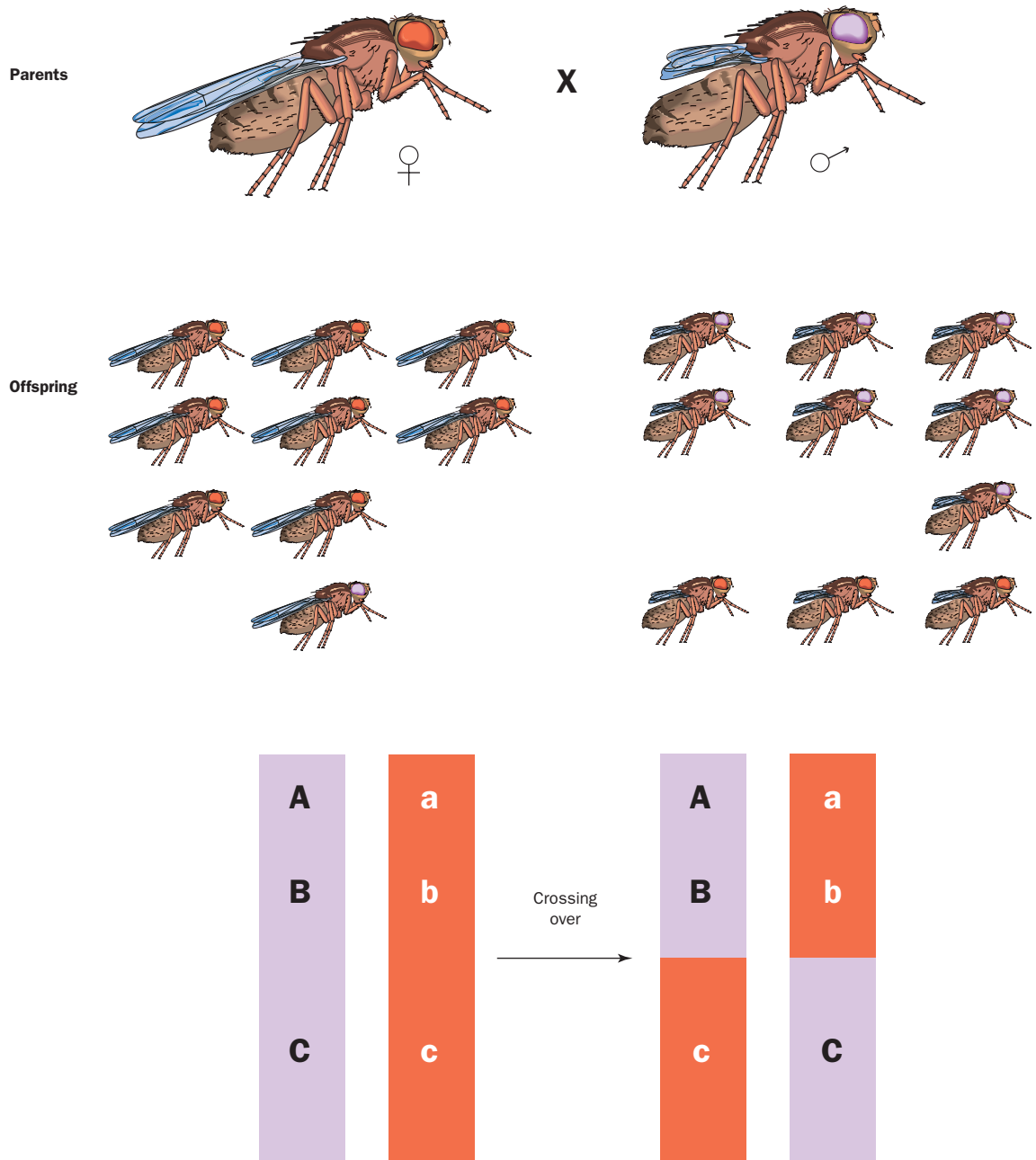


Thomas Hunt Morgan won the 1933 Nobel Prize in physiology or medicine for his discoveries concerning the role of chromosomes in heredity.

allele a particular form of a gene

recessive requiring the presence of two alleles to control the phenotype

LINKAGE AND CHROMOSOME MAPPING



A cross between two double heterozygotes indicates linkage. Very few recombinants (long wing-purple eye or short wing-red eye) are formed if the two genes are close together.

Morgan's results showed that the white-eye allele is inherited on the X chromosome, and confirmed the discovery that the X chromosome helps determine sex, first shown in 1905 by Sutton and Nettie Stevens. In one step, his discovery proved that genes, the factors governing inheritance, are carried on chromosomes, and that specific genes are carried on specific chromosomes. This provided the crucial evidence that genes are indeed discrete physical objects.

Linkage and Chromosome Mapping

The discovery of more mutated genes allowed Morgan's group to explore how genes are arranged on the chromosome, and to discover an exception to one of Mendel's laws of inheritance. Mendel had proposed the Law of Independent Assortment, stating that the alternative forms of different traits (such as round versus wrinkled pea seeds and short versus tall plant height) separate and recombine independently of each other, so that, for instance, obtaining a wrinkled tall plant is just as likely as obtaining a wrinkled short plant.

Morgan found this was not always true. Rather, certain combinations of alleles are very unlikely to be separated from each other, a fact he attributed to co-inheritance of the two alleles on the same chromosome. While alleles on separate chromosomes assort independently, as Mendel predicted, those on the same chromosome travel together unless separated.

To explore this, Morgan crossed a red-eyed fly with normal-length wings with a purple-eyed fly with stubby wings. After two generations, Mendel's laws predicted that all possible combinations of eye color and wing length should be equally likely. Instead, Morgan found that most flies had the original trait combinations, while red-eyed, stubby-winged flies were rare, as were purple-eyed, normal-winged flies. He concluded that the genes for wing length and purple eye color were on the same chromosome. Like passengers traveling on the same ship, once the particular alleles were together, they tended to stay linked. (Note that the purple eye-color gene is not the same one as the red-white eye-color gene he discovered previously, and is not on the X chromosome.)

However, Morgan noted specific allele combinations didn't always stay together: There were a few flies whose stubby-wing allele and purple-eye allele had become separated from each other. This led Morgan to propose that chromosomes sometimes exchange segments, allowing their passengers to change vessels, so to speak. This phenomenon is known as crossing over, and was later conclusively demonstrated in maize by Barbara McClintock.

Crossing over is now known to occur only during meiosis, the chromosome division that leads to formation of eggs and sperm. During meiosis, **homologous** chromosomes originally donated from the mother and father pair up for an extended period. In this period, called synapsis, the maternal and paternal chromosomes randomly exchange several segments, resulting in a pair of chromosomes with a mix of maternally derived and paternally derived alleles. These then separate to form the eggs and sperm.

Morgan's student Sturtevant reasoned that the likelihood of two alleles becoming separated during crossing over was proportional to the distance between them. In other words, the closer they are, the more likely they will stay together, and the further apart they are, the more likely they will separate. If A, B, and C are on the same chromosome, and A stays with B more often than it stays with C, then the distance from A to B is shorter than the distance from A to C. In this way, the relative distances of genes can be determined, providing a "linkage map" of the chromosomes. The unit of relative distance is called the morgan, in honor of Morgan himself. Calvin Bridges later devised a method to determine the absolute distance between genes, relying on the distinct banding patterns seen in *Drosophila* chromosomes in the larval stage.

homologous carrying similar genes



Morgan's Legacy

In 1915 Morgan, Bridges, Sturtevant, and Muller published *The Mechanism of Mendelian Heredity*, a highly influential textbook laying out the evidence for the chromosomal theory of heredity and illustrating their methods so others could apply them in further research. In 1928 Morgan moved to the California Institute of Technology to found the Division of Biology. Sturtevant and Bridges went with him. Five years later Morgan was awarded the Nobel Prize in physiology or medicine for his work in genetics. He shared the prize money with Sturtevant and Bridges. Besides his own discoveries, Morgan's intellectual legacy includes the historically important researchers who trained with him, including Theodosius Dobzhansky, who applied the new genetics to an understanding of evolution. Another of his students was George Beadle, who discovered that mutations affect the working of proteins, and proposed the "one gene—one enzyme" definition of the gene. SEE ALSO FRUIT FLY: *DROSOPHILA*; LINKAGE AND RECOMBINATION; McCLINTOCK, BARBARA; MEIOSIS; MENDEL, GREGOR; MULLER, HERMANN.

Richard Robinson

Bibliography

Allen, Garland E. *Thomas Hunt Morgan: The Man and His Science*. Princeton, NJ: Princeton University Press, 1978.

Judson, Horace F. *The Eighth Day of Creation: The Makers of the Revolution in Biology*. New York: Simon & Schuster, 1979.

Morgan, Thomas Hunt, et al. *The Mechanism of Mendelian Heredity*. New York: Holt Rinehart & Winston, 1915. Reprint, with an introduction by Garland E. Allen, New York: Johnson Reprint Corporation, 1978.

Sturtevant, Alfred H. *A History of Genetics*. New York: Harper & Row, 1965.

Mosaicism

In 1961 Mary Lyon, an English scientist, hypothesized that one of the two X chromosomes in females becomes genetically silent early in a female embryo's development. To understand how she arrived at this idea, which has come to be known as "the Lyon Hypothesis," we need to understand what was known about the sex chromosomes.

The Sex Chromosomes

Humans have twenty-three pairs of chromosomes, including one pair of sex chromosomes and twenty-two pairs of **autosomes**. The sex chromosomes are either X or Y chromosomes. Females have two X chromosomes, and males have an X and a Y chromosome.

In mammals, the sex of an individual is generally determined by whether the individual inherited an X or a Y chromosome from the father. The Y chromosome contains the *SRY* (Sex-determining Region Y) gene that directs male sexual development, but holds relatively few other genes. Many of the several dozen genes or gene families on the Y chromosome are necessary for the production of sperm. A handful are shared with the X chromosome, which is a medium-sized chromosome that is likely to contain more than one thousand genes.

autosomes chromo-
somes that are not sex-
determining (not X or Y)

Lyon knew that female mice that had only a single sex chromosome, the X chromosome, were normal. She also knew that mice carrying two different genes for coat color, one on each X chromosome, exhibited a mosaic, or blotchy, pattern of coat color. Some cells expressed one color gene, while others expressed the other, producing a mottled pattern.

Finally, she knew that when female cells are stained and looked at under a microscope a darkly staining region called a Barr body can be seen. She hypothesized that in female cells the Barr body is an inactive X chromosome. Thus, only one X chromosome would be active in any cell, resulting in a mottled pattern of X-linked gene expression. Furthermore, female cells lacking an X chromosome would be all right if the remaining X was the active one.

Mosaic Expression

A good example of an animal that exhibits mosaicism is a tortoiseshell cat, which has patches of black and orange fur. There is a dominant gene on the X chromosome that makes the cat's fur orange. If a female cat has this gene on only one of its two chromosomes, then the pigment-producing cells in which this chromosome is active will generate orange fur, while those that have the gene on the inactive X chromosome will make black fur.

The choice of which X chromosome to inactivate occurs very early in development, when an embryo has less than one hundred cells. While this initial choice is generally random, the same inactivation pattern is then passed on to descendant cells through subsequent cell divisions, resulting in a patch of cells with one or the other X chromosome active, and therefore producing orange or black fur in the tortoiseshell cat.

Because the single X chromosomes in males is never inactivated, male cats do not have tortoiseshell coats. XXY male cats, however, which have an extra X chromosome, can have such coats.

X Chromosome Inactivation

How does a cell manage to silence one X chromosome in a cell but not the other even though the two chromosomes are almost identical? A clue to this puzzle came from the discovery of a gene named *XIST* (X inactive specific transcripts). This is a gene that is expressed only on the inactive X chromosome. It is transcribed into an RNA that does not code for protein, unlike most genes. Instead, the RNA associates with the X chromosome from which it is made, resulting in silencing of the chromosome.

We know many of the components of this silencing process, and they are proteins that have been implicated in the silencing of other genes or regions of chromosomes as well. They are predominantly factors that influence the structure of the chromatin, which is the complex of DNA and proteins that is found in chromosomes. For example, the chromatin structure can be changed by adding methyl groups to the DNA, or by adding acetyl or methyl groups to the **histone** proteins with which the DNA interacts.

Effect of X Inactivation on Human Disease

Females with a mutated gene on an X chromosome have two populations of cells. One group produces the intact protein, and the other produces a protein that is affected by the mutation. Like tortoiseshell cats, these females

histone protein around which DNA winds in the chromosome